Automation Landscape Report: Cape Town Rooibos Primary-Processing Line

Current state on the ISA-95 levels, control architecture, data transport through load-shedding, and data integrity

Worked example by the DigiSkills programme team, for the Cape Town engineering and quality management teams

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Executive summary

Ubuntu Naturals' Cape Town plant dries, mills and sieves rooibos for Rooibos Recovery, which is packed and released in Mombasa. The line is instrumented from the field up to batch records, but those records do not yet hold together as evidence a GMP inspector would accept. This report maps what is on the floor onto the ISA-95 levels, recommends what should replace it and how its data should travel through load-shedding, and sets out what SAHPRA will expect.

The evidence is the site's own records: one week of one-minute meter readings (4 to 10 May 2026) and 30 days of pack-off batch records (April 2026). In that week the grid went down 4 times, for 600 minutes, 480 of them in shift hours; the longest outage lasted 240 minutes. The meter's logger lost 360 minutes of readings in one overnight gap and wrote one minute twice. For 60 minutes it recorded values up to 1,099 kW, 3.6 times the running load, beside an unsigned note. The batch records name no operator, and their energy field cannot be reconciled with the meter.

None of this stops production. All of it would be found in a data integrity inspection, because each fault breaks an ALCOA+ principle that a validated system is expected to enforce.

The report recommends four decisions.

1. Keep PLC control, one controller per process area under a single site SCADA, rather than buying a DCS. The line is small and largely sequential, and the site's technicians already maintain PLCs.
2. Install an edge historian with store-and-forward on the UPS and the genset's critical circuit, so that neither a grid nor a WAN outage can lose a record.
3. Use OPC UA inside the plant and MQTT with Sparkplug B over TLS to the group MES in Johannesburg, keeping Modbus only behind a gateway for the power meters.
4. Validate the system under GAMP 5, with audit trails, named user log-ins and one synchronised clock in place before the first record that counts as GMP evidence.

Section 8 turns these into actions for engineering and for quality, and Section 9 phases the work so that the line keeps producing.

1 Introduction and scope

1.1 The line

Cape Town is Ubuntu Naturals' Tier 1 primary-processing plant in South Africa, at Stage 2 of the group's value chain. Rooibos arrives from Western Cape smallholder farmer-members. It is dried in a gas-assisted electric drying tunnel, hammer-milled, sieved, and conveyed pneumatically to a pack-off filler that doses the milled material by volume into packs for Mombasa. Dust extraction, compressed air, HVAC and lighting make up the rest of the electrical load.

The drying tunnel matters most to this report. It is the line's largest electrical load and the one process that cannot simply stop when the grid fails: a tunnel full of part-dried leaf has to be held in a safe state, so the diesel genset carries the tunnel hold and the dust extraction during an outage.

1.2 Why this matters to quality

The milled rooibos is the herbal starting material for Rooibos Recovery. Its drying and milling records become part of the batch history that the quality unit in Mombasa reviews before release, so a gap in Cape Town's records is a gap in the finished product's record. As a South African manufacturing site, Cape Town falls under SAHPRA's GMP inspections, and the product's African export markets may expect the same evidence as the AMRH harmonisation work progresses. Section 7 sets out what those expectations mean for the automation.

1.3 Scope

The report covers the line from raw-material intake to pack-off, with the utilities that serve it. It does not cover the group ERP or MES themselves, which are owned centrally, or the release decision in Mombasa, but it does cover Cape Town's interfaces to all three.

1.4 Evidence and method

Two sets of records were analysed. The whole-site meter export gives the load, the ambient temperature and a status label for every minute of one week. The pack-off batch export gives one row per batch with a fill volume, a temperature, an energy reading and the shift. Every figure in this report was computed from those files by a script, so any reader can check it. Descriptions of controllers, panels and networks come from the site survey, which the records cannot confirm. Scores, weights, sizing estimates and the architecture are the author's design choices. The programme lectures L1, L2, L6, L8 and L11 are the reference frame: L2 for ALCOA+ and data integrity in computerised systems, and L11 for the regulatory framework and 21 CFR Part 11.

2 What the site's records show

Table 1. Findings from the site's own records, and what each says about the systems that wrote them.

Finding	Evidence from the records	What it says about the systems
Grid outages	4 load-shed blocks, 600 minutes in total, 480 of them in shift hours; longest block 240 minutes	Control, records and network must ride through a four-hour outage as a routine event, not an emergency.
Genset load	Mean 48.1 kW during outages, against 49.4 kW at overnight idle; 481 kWh from diesel in the week	The critical circuit carries about the overnight base load. The control system's own supply must be on it.
Restart	4 restarts, peaking at 518 kW in one minute against a running mean of 302.8 kW	Every machine restarts at once. Nothing on the line sequences the restart.
Logger gap	9,720 distinct minutes logged out of 10,080, a coverage of 96.4%; one overnight gap of 360 minutes	The logger writes to a local SD card with no buffer and no alarm. The loss shows only when the rows are counted.
Duplicate	1 minute written twice: 9,721 rows for 9,720 minutes	The record has no unique key, so a re-write cannot be told apart from a second reading.
Metering fault	60 minutes questioned in a free-text note, peaking at 1,099 kW	Doubtful values stay in the record with no reason code, no author and no review.
Batch records	5,400 records over 30 days, one every 8 minutes, 7 fields	No operator, no reviewer, and no unit or location for the temperature field.
Shift pattern	1,800 Night-shift batch records in April; 0 running minutes before 05:00 in the May week	The two systems disagree about when the line runs.
Energy	Batch energy averages 307.6 kWh a day; the meter logged 5,901 kWh on Monday 4 May	Neither record defines what its meter covers.
Fill volume	7 of 5,400 doses outside the 490 to 510 ml limits	The measurement is captured; its disposition is not.

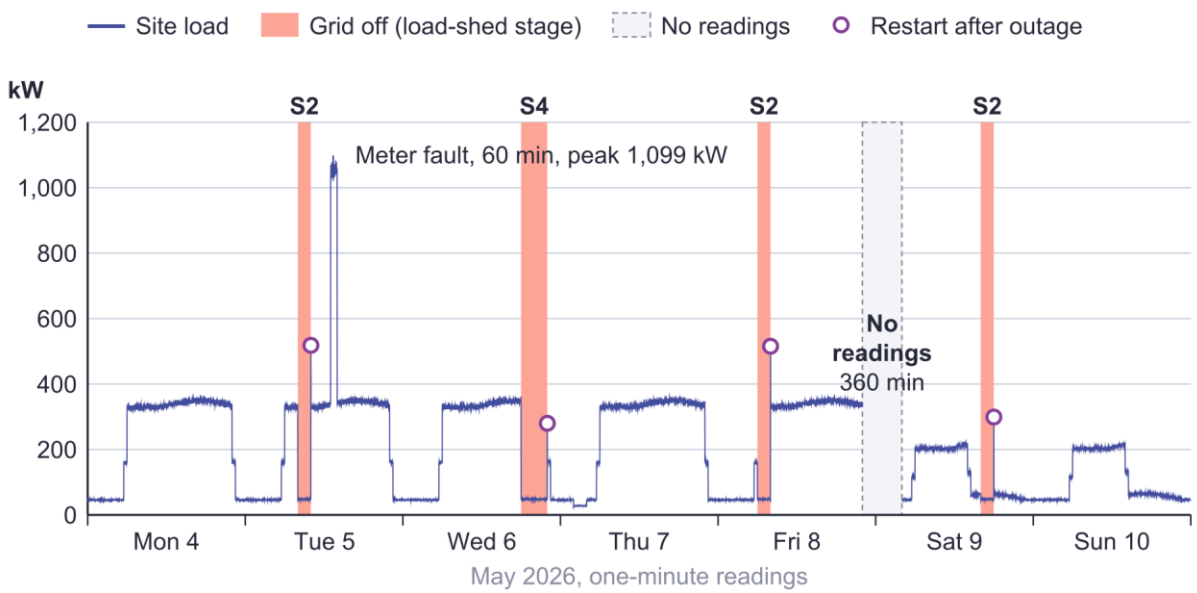


Figure 1. The site meter over the week of 4 May 2026. Shaded bands are load-shedding blocks (S2 and S4 are the Eskom stages the status notes record). The grey band is the logger gap. Circles mark the restart minute after each block.

Figure 1 shows the expected pattern: two weekday shifts from 06:00 to 22:00, a short weekend shift, a base load of about 49.4 kW overnight, and four load-shedding blocks, each ending in a restart spike.

Three features are faults in the record rather than in the process. The six-hour hole on Friday night is the logger, not the plant, and the file gives no sign of it beyond the missing timestamps. It is 1.5 times as long as the longest outage, so the logger loses more data on its own than load-shedding takes from it. The Tuesday spike is a metering fault that someone noticed, but the readings were left in place behind an unsigned note, so a daily total includes them unless the analyst knows to look. The duplicate minute on Wednesday is invisible on the chart and is counted twice by anyone who sums the column.

The outages also say something about control. The genset's mean load during an outage (48.1 kW) is close to the overnight base load (49.4 kW), so the critical circuit carries the tunnel hold and dust extraction and little else. When the grid returned, the plant drew up to 518 kW in one minute, 1.7 times the running mean of 302.8 kW: every machine restarting at once. A sequenced restart is a control function, and no system on the line owns it today.

The batch records are clean in the sense that matters to a spreadsheet: 5,400 rows, one every 8 minutes for 30 days, with no gaps. But none of their 7 fields says who made the batch, who reviewed it, or what the temperature column measures. They place 1,800 batches on a Night shift in April, while the meter shows the line idle before 05:00 throughout the May week. Either the shift pattern changed, or one system is not timestamping production events. The batch energy field adds up to 307.6 kWh a day, while the meter logged 5,901 kWh on a normal Monday, about 19 times as much, and neither file says what its meter covers. Two systems on one line share no definitions.

3 The ISA-95 levels and where the line sits

3.1 The five levels

ISA-95, published internationally as IEC 62264, defines levels by the decisions each one makes and the time scale it works in, not by the box that does the work. A single panel can hold Level 1 control and a Level 2 display. The value of the model here is that it shows where information has to cross a boundary, because that is where records are lost, altered or never made.

Table 2. The ISA-95 levels, defined with examples from this line.

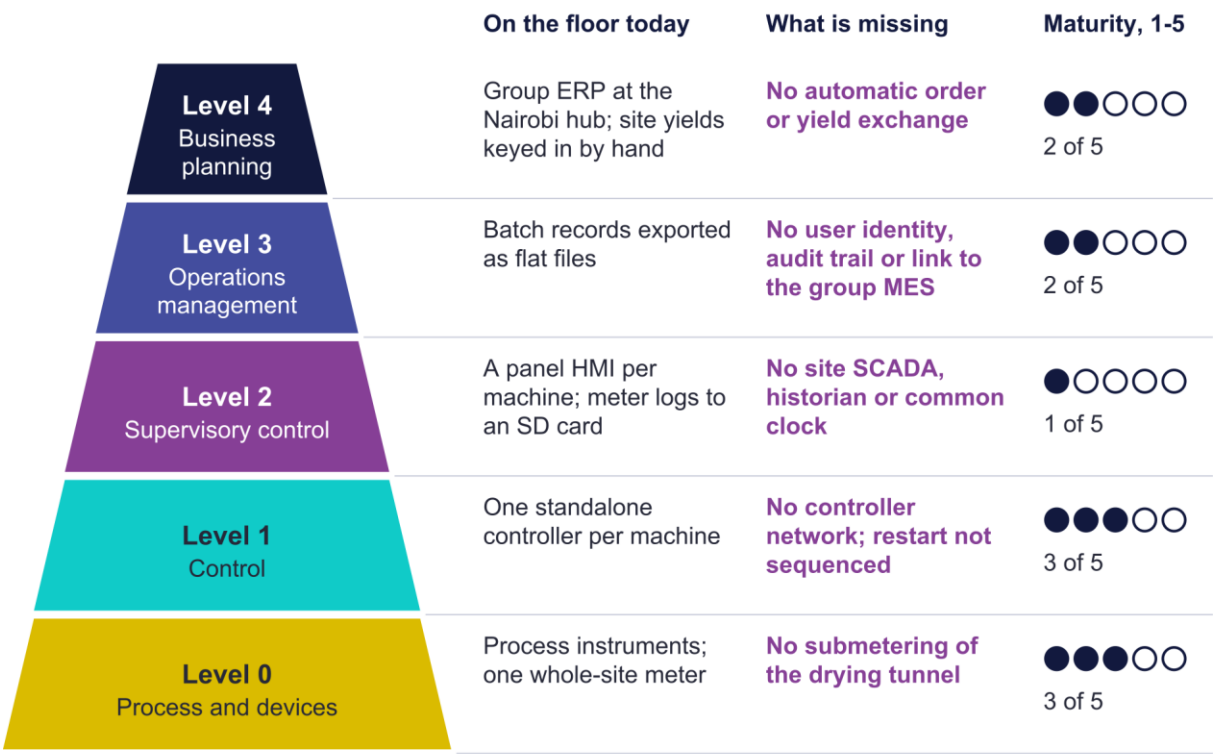
Level	What it does	On the Cape Town line
0 Physical process	The material and the equipment that transforms it	Leaf moving through the drying tunnel, the mill and the sieves; the burner, motors and fans
1 Sensing and manipulation	Instruments and the controllers that act on them, in milliseconds to seconds	The tunnel air-temperature probe and the controller that trims the burner; the filler's dose control
2 Monitoring and supervisory control	Displays, alarms, trends and set points across a process area, in seconds to minutes	An operator raising the tunnel set point for a wetter intake lot and reviewing the alarm history
3 Manufacturing operations management	Batch execution, records, quality, maintenance and performance, in minutes to days	The batch record for a lot of milled rooibos: drying profile, yield, deviations and review
4 Business planning and logistics	Orders, inventory and costing, in days to months	The ERP order that sends a lot to Mombasa and books its cost

3.2 Current state and gaps

The group describes Cape Town as instrumented from Level 0 to Level 3, and it is. There are sensors, controllers, a site meter and batch records. What the records in Section 2 show is that these do not yet work as one system. Table 3 places each system on its level, names what is missing, and says why the gap matters at a SAHPRA inspection. Figure 2 shows the same picture as a pyramid, with a maturity score per level.

Table 3. Current state and gaps by ISA-95 level. Maturity runs from 1 (absent or manual) to 5 (integrated, validated and used for improvement) and is the author's assessment.

Level	On the floor today	What is missing	Why an inspector would care	Maturity
4	Group ERP at the Nairobi hub; site yields keyed in by hand	No automatic order or yield exchange	A hand-keyed copy between the batch record and the ERP is an unverified transcription.	2
3	Batch records exported as flat files	No user identity, audit trail or link to the group MES	The batch record is the GMP record. Without identity and an audit trail it is neither attributable nor reviewable.	2
2	A panel HMI per machine; meter logs to an SD card	No site SCADA, historian or common clock	Process data that supports a batch decision must be complete and consistent in time. Section 2 shows it is neither.	1
1	One standalone controller per machine	No controller network; restart not sequenced	An unrecorded change to controller logic changes a validated process without a trace.	3
0	Process instruments; one whole-site meter	No submetering of the drying tunnel	Tunnel air temperature is likely to be a critical process parameter, so its instrument must be calibrated and traceable.	3



Maturity: 1 absent or manual, 3 reliable but isolated, 5 integrated and validated. Author's assessment.

Figure 2. The ISA-95 levels at Cape Town: what is on the floor, what is missing, and the author's maturity score for each level.

The most serious gaps are at Levels 2 and 3, and they are linked. Level 2 has no site SCADA, historian or common clock, so process data lives on machine panels and an SD card. Level 3 has batch records but no user identity, no audit trail and no link to the group MES in Johannesburg. An inspector who asks for the drying profile behind a released lot will be shown a file that cannot prove who made it, when, or whether it has changed. Level 1 runs the process reliably; its weakness is isolation, with no shared network or clock and no record of logic changes. Level 0 lacks submetering, so the tunnel's share of the load can only be inferred.

3.3 Closing the gaps in order

The gaps cannot be closed in any order, because some capabilities depend on others.

1. One clock and named users (Levels 2 and 3), because every audit trail installed later depends on them.
2. An edge historian with store-and-forward (Level 2). It stops records being lost now, before any controller is touched.
3. A site SCADA over a controller network (Levels 1 and 2), which needs the clock and named users and feeds the historian.
4. Controller upgrades (Level 1), one area at a time, with the sequenced restart and requalification of each area.
5. The interface to the group MES and electronic batch records (Level 3), worth validating only once Level 2 data can be trusted.
6. Automatic exchange with the ERP (Level 4), last because it adds least to GMP compliance.

4 PLC or DCS for the upgrade

4.1 The comparison

A programmable logic controller (PLC) is a rugged computer that runs control logic for a machine or process area, normally supervised by a separate SCADA package. A distributed control system (DCS) is a single-vendor platform in which controllers, operator stations, historian and engineering tools share one database. Table 4 compares the two against the criteria that decide the choice here.

Table 4. PLC with SCADA compared with a DCS for the Cape Town line.

Criterion	PLC with site SCADA	DCS	What it means at Cape Town
Scalability	Add a controller per area; SCADA licensing grows with signal count	Thousands of loops under one database and one engineering tool	Four process areas and a few hundred signals are well inside PLC scale.
Capital cost	Lower hardware and licence cost per input or output; integration engineering on top	Higher entry cost, with engineering tools built in	DCS entry cost would be spread over very few loops.
GxP suitability	Audit trail, electronic signatures and batch functions are available, but the integrator must configure and validate them	Audit trail, signatures and ISA-88 batch functions are native and consistent	Both can be validated. A DCS is easier to validate; PLC and SCADA are easier to staff.
Vendor lock-in	Moderate: OPC UA lets controllers and SCADA come from different vendors	High: controllers, I/O, software and spares come from one vendor for the plant's life	A single-vendor DCS ties a Tier 1 site to one service channel in the Western Cape.
Maintenance complexity	Familiar to electricians and technicians; logic in IEC 61131-3 languages	Needs DCS-trained engineers; changes through the vendor's engineering station	The site's skills favour PLCs.
Connectivity	OPC UA, MQTT and Modbus are common on current PLCs and SCADA	OPC UA standard; other links through gateways or vendor modules	Either can reach the group MES.
Process fit	Strong for sequences and discrete control; adequate for a handful of PID loops	Strong for many interacting continuous loops	The tunnel has a few loops; most of the line is sequence logic.
Power-loss behaviour	Restart behaviour is set per program and must be designed deliberately	Redundant controllers and supplies are standard; restart is managed system-wide	Load-shedding makes restart design central either way.

4.2 Recommendation

The recommendation is PLC control under a single site SCADA: one modern PLC per process area (drying tunnel, milling and sieving, conveying and pack-off, utilities), networked on a controller ring, supervised by a redundant pair of SCADA servers, with an edge historian beside them.

Four constraints decide it. First, the standalone controllers can be replaced one area at a time in idle windows, whereas a DCS gives most of its value only when the whole plant is on it. Second, the site's technicians maintain PLCs today, and PLC skills and spares are available locally in the Western Cape. Third, GxP compliance does not require a DCS: a validated, configured SCADA and historian provide audit trails, electronic signatures and access control. Fourth, load-shedding puts the same demands on either architecture. What protects the record is the UPS, the restart logic and the edge buffer, and none of those comes from choosing a DCS.

4.3 The hybrid option

PLC-based process control systems, such as Rockwell's PlantPAx or Siemens' PCS 7, put PLC-class controllers under a DCS-style engineering environment with a shared database, an ISA-88

batch engine and a built-in audit trail. They are the realistic alternative and the one to price if the site grows. Cape Town is multi-botanical; if drying lines are added for other botanicals, a shared database and batch engine begin to pay for themselves, so the PLC and SCADA platforms should be chosen now to grow into one without replacing the controllers.

4.4 Total cost of ownership

Without vendor quotations the comparison can only be made by cost element, and Table 5 does that. The DCS saves engineering and validation effort. The PLC option saves on hardware, spares and people. For a line of this size the people and spares costs recur every year, while the engineering savings occur once, so over a ten-year life the comparison favours PLC and SCADA. This is the author's reasoning, not a costed estimate, and the site should obtain quotations for both before capital is approved.

Table 5. Total cost of ownership by cost element (author's qualitative assessment).

Cost element	PLC with site SCADA	DCS
Hardware and licences	Lower	Higher
Engineering and integration	Higher: the integrator configures audit trail, batch and alarm handling	Lower for the same functions; each change goes through the vendor's tools
Validation	Higher: more interfaces to test	Lower: native functions come with vendor documentation
Spares and support	Lower: several local sources	Higher: one vendor, often imported spares
Staff and training	Lower: existing skills	Higher: new skills or a service contract
Cost of a load-shedding block	Set by UPS, genset and restart design, not by architecture	The same

4.5 Migration path

The controllers move one area at a time. Utilities go first, because PLC 4 carries the transfer-switch and genset signals that the restart sequence on every other area needs. The drying tunnel follows, then milling and sieving, then conveying and pack-off. Each cut-over fits an idle window; Section 9 shows that the line is idle for 3,329 minutes in a week. The old controller's logic is documented before it is removed, so that the new program can be tested against the behaviour it replaces.

5 Data transport protocols

5.1 What the plant needs from a protocol

Five criteria follow from Section 2. Outages are the most frequent threat to the record, so surviving one carries the most weight. The audit trail is what an inspector will test, so it comes next. Security matters because the same links will reach Johannesburg. Bandwidth matters most on the WAN, which is the narrowest link. Edge processing and the size of the supplier ecosystem matter least. The weights in Table 6 are the author's.

5.2 The four protocols

Modbus, as RTU on serial lines or TCP on Ethernet, is a simple register protocol that nearly every power meter and drive supports. It carries no timestamps, quality codes or data types beyond registers, and its common forms have no authentication or encryption. The poller stamps each reading on arrival, which breaks the Contemporaneous principle whenever there is a delay, and there is no store-and-forward. For GxP use, the first system that reads a Modbus value must timestamp and store it.

OPC UA (IEC 62541) carries every value with source and server timestamps and a quality code, inside a typed information model that can describe the equipment hierarchy. It authenticates applications with X.509 certificates and users with accounts, and can sign and encrypt every message. Subscriptions survive short interruptions but not a four-hour outage, so history across

a long gap needs a historian. For GxP it is the strongest of the four, because the timestamp and quality travel with the value.

MQTT (an OASIS standard, now version 5.0) is a publish-subscribe protocol in which clients send messages to a broker. It is light on bandwidth and reports by exception. Persistent sessions and QoS 1 let a client hold unacknowledged messages and resend them when the link returns, which is the basis of store-and-forward. Sparkplug B adds a defined payload with timestamps, sequence numbers, and birth and death certificates that tell the host when an edge node has gone offline. Security comes from TLS and broker access control lists. For GxP, the sequence numbers make a gap detectable.

WebSockets (RFC 6455) give a browser a two-way channel over HTTP, which suits live dashboards. They define no data model, no timestamps and no delivery guarantee across a reconnection. They belong at the top of the stack, for viewing data, not for moving GMP records.

Table 6. Protocol decision matrix. Weights and scores (1 poor to 5 strong) are the author's; the weighted score is computed from them.

Criterion (weight)	Modbus TCP	OPC UA	MQTT with Sparkplug B	WebSockets
Survives an outage (store-and-forward, session recovery) (30)	1	3	5	2
Supports the GxP audit trail (timestamp, quality, identity) (25)	1	5	3	1
Security (authentication, encryption) (20)	1	5	4	3
Bandwidth on the WAN to Johannesburg (15)	3	2	5	3
Edge processing and ecosystem (10)	3	4	5	2
Weighted score out of 100	**30**	**77**	**86**	**42**

MQTT with Sparkplug B scores highest on the plant's priorities, and OPC UA second. They do not compete here. Inside the plant the network is local, short and on UPS, bandwidth is not a constraint, and OPC UA's timestamps, quality codes and certificates do most for the record. Across the WAN to Johannesburg the link is narrow and an outage is likely, and MQTT's report-by-exception and resend on reconnection do most. Modbus stays only where a device offers nothing else, behind a gateway that timestamps and stores each reading as it takes it. WebSockets serve dashboards only.

5.3 Store-and-forward and the edge buffer

The design is sized from the week's records. The longest grid outage was 240 minutes and the longest loss of readings 360 minutes. The edge historian keeps 7 days locally, a design choice, so the longest outage would use 2.4% of its capacity. At an estimated 400 signals sampled once a second and about 16 bytes per stored value before compression, 7 days of history is about 3.9 GB, which one industrial edge server holds comfortably. These sizing figures are the author's estimates, to be replaced from the site's I/O list.

The mechanism has five rules.

1. Every value is timestamped by the controller or edge node that reads it, from the site time server, never by the system that receives it later.

2. The edge node writes each value to local storage, with a sequence number, before it publishes it.

3. A value is deleted locally only after the group broker acknowledges it (MQTT QoS 1 over a persistent session).

4. On reconnection the edge sends alarms and events first, then batch records, then process history, oldest first within each class, at a throttled rate so that the backlog does not crowd out live data.

5. The group MES checks the sequence numbers. A batch with a gap is marked incomplete and cannot be released until the gap is explained.

The edge node, the controller network switches, the time server and the SCADA servers sit on a UPS with 30 minutes of autonomy, which bridges the transfer to the genset, and on the genset's critical circuit. The one-minute records cannot show how long the transfer takes, so the UPS must be sized from the genset's own transfer specification, not from this data.

5.4 Protocol security and staged migration

Each protocol brings its own security risk. Modbus accepts a write from anything that can reach it, so it is confined to one gateway segment with no route upward. OPC UA runs in Sign and Encrypt mode with site-issued certificates, and the edge historian connects as a read-only client. MQTT runs over TLS on one outbound port, with a broker account per site that can publish only under Cape Town's own namespace.

The migration follows the phases in Section 9. First, the edge node and time server read the existing controllers and meter, read-only. Second, each area moves to OPC UA as its controller is replaced, and its Modbus mapping is retired. Third, the edge connects to the group broker and the interface is validated. Last, every remaining Modbus path is removed except the power meters' serial links behind the gateway.

6 Network architecture

6.1 Reading the diagram

Figure 3 runs from the field devices at the bottom to the group ERP at the top. The four PLCs share a controller ring, so one cable or switch failure cuts none of them off, and the SCADA servers run as a hot-standby pair. The edge historian reads the controllers through OPC UA and is the only site system that exchanges data with the group, through the DMZ, as MQTT over TLS. The WAN has an LTE backup because a local exchange can lose power during load-shedding even when the plant does not. Elements with a dashed amber outline are on the UPS and the genset's critical circuit.

6.2 What happens in a load-shedding block

At the start of a block the automatic transfer switch moves the critical circuit to the genset, and the UPS carries the control network, the tunnel and utilities controllers, the servers, the time server and the edge node through the transfer. PLC 4 signals the outage. PLC 1 moves the drying tunnel to its hold state, and PLCs 2 and 3 stop their areas in sequence and record the stop as an event. If the WAN fails too, the edge node keeps writing locally, and SCADA keeps working because nothing in it depends on the WAN.

When the grid returns, PLC 4 waits for a stable supply before it permits a restart, and the areas restart in sequence (utilities, the tunnel, milling, then pack-off) rather than together. That should remove most of the 518 kW one-minute spike seen in the records. The edge node then empties its buffer in the order set in Section 5.3.

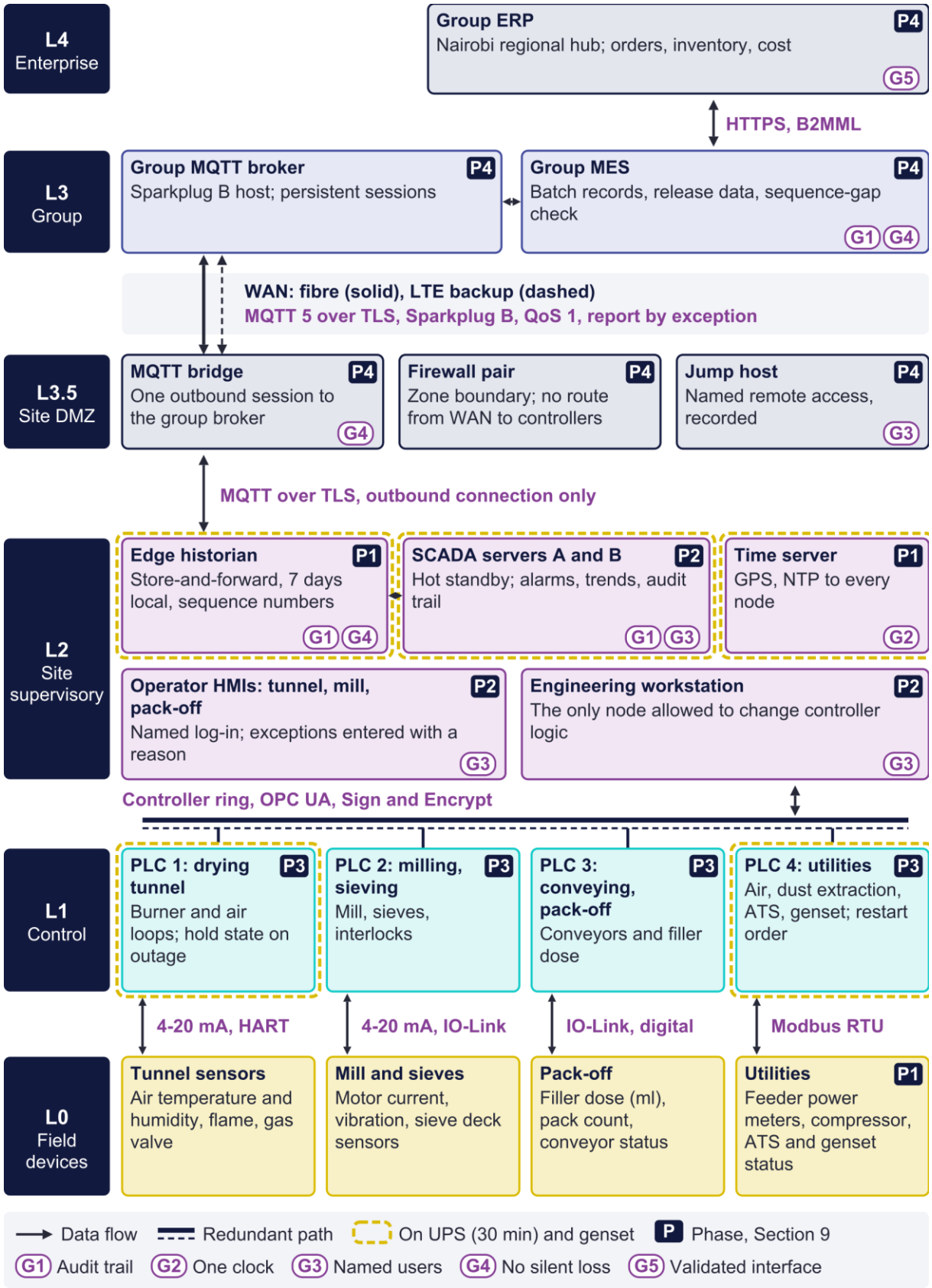


Figure 3. Network architecture for the Cape Town line, from field devices to the group ERP. Protocols are labelled in purple at each boundary. G tags mark data integrity controls; P tags mark the commissioning phase in which each element is installed.

6.3 Zones and conduits

The architecture follows the Purdue model's segmentation, with security zones and conduits in the sense of IEC 62443. Table 7 lists them. The principle is that nothing outside the site can open a connection to a controller.

Table 7. Security zones and conduits (IEC 62443).

Zone	Contents	Conduit	Controls
Enterprise	Group ERP	To group operations	HTTPS with mutual TLS; group IT controls
Group operations	Group MES, MQTT broker, platform	To the site DMZ over the WAN	MQTT over TLS on one port; broker access lists per site
Site DMZ	Firewall pair, MQTT bridge, jump host	Outbound to the group; inbound to site supervisory	No route from the WAN to controllers; remote access only through the jump host, with named accounts and recorded sessions
Site supervisory	SCADA, historian, HMIs, time server, engineering workstation	To the control zone	OPC UA Sign and Encrypt; only the engineering workstation may change controller logic
Control	PLCs and field devices	To site supervisory only	No route beyond Level 2; Modbus confined to the gateway segment

7 Regulation and data integrity

7.1 The framework

SAHPRA inspects South African manufacturers against its Guide to Good Manufacturing Practice, which follows the PIC/S GMP Guide. Two of its annexes bear on this line: Annex 7, on herbal medicinal products, and Annex 11, on computerised systems. Annex 11 expects a computerised system to be validated for its intended use, to keep an audit trail of GMP-relevant changes and deletions, to control access, and to keep data secure and retrievable for its retention period. The WHO guideline on data integrity (Technical Report Series 1033, Annex 4) and PIC/S PI 041 set out ALCOA+ and how inspectors test it. 21 CFR Part 11, the United States rule on electronic records and signatures, does not govern Cape Town directly, but lecture L11 uses it as the reference for electronic signatures.

The African Medicines Regulatory Harmonisation programme and the African Medicines Agency aim to harmonise GMP expectations and to support regulators relying on one another's inspections, so evidence that satisfies SAHPRA is evidence other African regulators may also ask to see. Mombasa, where Rooibos Recovery is packed and released, answers to Kenya's Pharmacy and Poisons Board, which can be expected to ask for the Cape Town records that come with each lot.

7.2 ALCOA+ against the line

Table 8 takes each ALCOA+ principle in turn: the gap the records show today, the capability in the proposed system that closes it, an implementation example, and the compliance risk before and after. The risk ratings are the author's.

Table 8. ALCOA+ principles: current gaps, proposed capabilities and compliance risk.

Principle	Gap today (evidence)	Proposed capability	Implementation example	Risk now / after
Attributable	Batch records name no operator; the metering-fault note has no author	Named log-in on every HMI; every entry and change recorded against a user	Badge and PIN at the pack-off HMI; no shared operator account	High / low
Legible	A temperature field with no unit or location; status codes undefined	Tag names, units and descriptions from one equipment model	Tunnel air temperature held as a named tag with its unit and instrument number	Medium / low

Principle	Gap today (evidence)	Proposed capability	Implementation example	Risk now / after
Contemporaneous	Batch timestamps disagree with the meter about when the line runs	Every value stamped at source from one GPS-disciplined clock	NTP to every controller and the edge node; an alarm if any clock drifts by more than a second	High / low
Original	1 minute written twice, indistinguishable from a second reading	The historian keeps the first value; re-writes are rejected and logged	A unique key on tag and timestamp; duplicate attempts written to the audit trail	Medium / low
Accurate	60 minutes of implausible readings left in place	Range and rate checks at the edge; exclusion only through a reviewed workflow	A reading above the feeder's rating gets a bad-quality code; a person excludes it with a reason and the original is kept	High / low
Complete	A 360-minute gap, found only by counting rows	Store-and-forward with sequence numbers; any gap raises an alarm	A Sparkplug sequence gap marks the batch incomplete in the MES	High / low
Consistent	Batch energy and site energy cannot be reconciled	One equipment and meter model shared by SCADA, historian and MES	Each meter declares the loads it covers; a daily reconciliation report	Medium / low
Enduring	Readings held on an SD card in the logger	Records on redundant servers, backed up and retained for the product's retention period	Daily backup to the group platform with a tested restore	High / low
Available	Records exported to files by hand	Records retrievable on request for review and inspection	An inspector's query answered from the historian and MES during the inspection	Medium / low

7.3 Cybersecurity as part of data integrity

Lecture L2 treats cybersecurity and data integrity as one subject, and so does the design. Access is role-based: operators cannot change set points outside validated ranges, supervisors cannot edit logic, and only administrators outside production manage accounts. No production role can switch off or edit an audit trail, and audit-trail review is part of batch review. Backups are tested by restoring them. Patches arrive through the DMZ and are tested before they reach the supervisory zone, and every remote session through the jump host is recorded. A cyber incident that changes a record is a data integrity event and goes through the same deviation process.

7.4 Validation outline

The new system is validated under GAMP 5 (Second Edition). The SCADA, historian and MES interface are configured products (Category 4). The controller programs are written for this line and are treated as custom applications (Category 5). The network and servers are infrastructure (Category 1). The outline of the validation master plan follows.

1. User requirements, including the data integrity controls marked G1 to G5 in Figure 3.
2. A risk assessment that ranks each function by its impact on product quality and on the record.
3. Functional and design specifications, with design qualification against the requirements.
4. Factory and site acceptance tests with the suppliers.
5. Installation qualification of hardware, software versions and network configuration.

6. Operational qualification, including a planned outage test: the grid supply is cut in an idle window, and the test proves that no record is lost, the sequence numbers are continuous, and the restart runs in order.
7. Performance qualification over normal production, with batches reviewed from the electronic record.
8. Periodic review, with change control for every later modification.

8 Recommendations

Each recommendation has an engineering action and a quality action, so that each team can act on it without the other translating. The phases are those of Section 9.

Table 9. Recommendations with engineering and quality actions.

No.	Recommendation	Engineering action	Quality action	Phase
R1	One site clock	Install a GPS time server; synchronise every controller, HMI, server and the edge node; alarm on drift	Add clock synchronisation to the user requirements and test it in OQ	P1
R2	Named users	Create named accounts on every HMI and SCADA client; remove shared log-ins	Approve the role matrix; review access quarterly	P1
R3	Edge historian with store-and-forward	Install it on the UPS and the critical circuit; connect read-only to the existing controllers and meter	Make the historian the GMP record for process data; write the procedure for reviewing gap alarms	P1
R4	Submeter the tunnel and the other feeders	Fit feeder meters on Modbus RTU behind the gateway	Record in the equipment model which loads each meter covers	P1
R5	Redundant site SCADA	Install the server pair and the controller ring	Validate the audit trail, alarm and electronic-signature configuration	P2
R6	Exception workflow for doubtful data	Add range and rate checks with quality codes at the edge	Write the procedure for reviewed exclusion with a reason, keeping the original	P2
R7	Replace controllers area by area, with a sequenced restart	Utilities, then the tunnel, milling and pack-off, each in an idle window	Change control and requalification for each area	P3
R8	Interface to the group MES	Connect the edge to the group broker over MQTT with TLS through the DMZ	Validate the interface; adopt the incomplete-batch rule	P4

9 Phased commissioning

The line must keep producing while the work is done, and the meter records show when it can be touched. In the May week the line was idle for 3,329 minutes, mostly on weekday nights after 22:00 and on weekend afternoons. The planned burner clean (120 minutes early on Thursday) shows that the site already schedules work at night. Every phase uses those windows.

Table 10. Commissioning phases.

Phase	Scope	Effect on production	Exit criterion
P1 Records first	Time server, named users, edge historian, submeters; all connections read-only	None: connections made in idle windows	A full week of records with no unexplained gap
P2 Supervisory layer	SCADA pair, operator HMIs, controller ring, exception workflow	Old panels stay live while operators train on the new HMIs	Operational qualification passed, including the planned outage test
P3 Controllers, area by area	Four PLC replacements with the sequenced restart	One area per weekend window, requalified before Monday	Each area’s first batch reviewed and released

Phase	Scope	Effect on production	Exit criterion
P4 Group integration	MQTT interface to the group MES, electronic batch records, ERP exchange	None at the line	Interface validated; first batch released on electronic records

10 Limitations

The records cover one week in May and one month in April, so seasonal effects, including winter load-shedding patterns and summer drying loads, are not represented. The controller inventory comes from the site survey, not from the records. Scores, weights, sizing and the cost comparison are the author's judgement and should be replaced by vendor quotations and the site's own I/O list. One-minute data cannot show events shorter than a minute, including the genset transfer itself. Finally, the report has not benchmarked the design against published inspection findings at comparable African facilities, which a full engagement would do.

11 Glossary

Term	Meaning
ALCOA+	Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring and Available: the qualities a GMP record must have
AMRH	African Medicines Regulatory Harmonisation, the programme aligning African medicines regulators
ATS	Automatic transfer switch, which moves loads between grid and genset
B2MML	Business To Manufacturing Markup Language, an XML form of the ISA-95 messages
DCS	Distributed control system
DMZ	Demilitarised zone: a network segment between the plant and outside networks
ERP	Enterprise resource planning (Level 4)
GAMP 5	ISPE's guide to a risk-based approach to computerised system validation
GMP, GxP	Good manufacturing practice; GxP covers all the regulated good practices
HMI	Human-machine interface: an operator's screen
IQ, OQ, PQ	Installation, operational and performance qualification
MES	Manufacturing execution system (Level 3)
MQTT	A publish-subscribe messaging protocol; Sparkplug B defines its payload for industry
NTP	Network Time Protocol
OPC UA	OPC Unified Architecture, an industrial data exchange standard (IEC 62541)
PIC/S	Pharmaceutical Inspection Co-operation Scheme
PLC	Programmable logic controller
QoS	Quality of service: MQTT's delivery guarantee level
SAHPRA	South African Health Products Regulatory Authority
SCADA	Supervisory control and data acquisition (Level 2)
TLS	Transport Layer Security
UPS	Uninterruptible power supply

12 References

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